Full keyboard - Revision H engineering audit

Updated 21 September 2026. MANUFACTURING HOLD. This is the current engineering work set, not an approved fabrication package. Revision G is superseded for design review and must not be built. No physical prototype has been tested. Read `OPEN-ITEMS.md` and `CHECK-RESULTS.md` before using any fabrication exports. A replacement manufacturer ZIP has not been issued.

See USB-C and headphones for port selection, charging limits, future programming and the two independent Bluetooth roles.

Work to quote and deliver

Please quote **one assembled engineering prototype**, with PCB/component minimum-order quantities and non-recurring engineering charges shown separately. PCBWay or another specialist may quote fabrication, assembly and enclosure manufacture. Retain the JLCPCB catalog references for chips and passives; purchase by the exact manufacturer part number and full specification. Quote enclosure manufacture, wire harnesses, final assembly, programming and the acceptance tests in this package. Return any necessary engineering changes in editable native files with regenerated manufacturing exports.

This work set may support an engineering review/quotation only. Before ordering, close the open design items and complete the explicit pre-build items in `assembly/ACCEPTANCE.md`: electrical-rating review, differential-impedance/stackup approval, procurement specifications, battery documentation and a small mechanical fit sample. After assembly, qualify the complete system against that test plan. These are manufacturer tasks in the requested prototype build, not claims that physical testing has already occurred. **Do not start a volume production run from this archive without a passed prototype and approved change record.**

Files to use

Folder/file	Purpose
`kicad/`	Eight KiCad 9 projects, schematics, routed boards, portable local footprints and 3D models
`manufacturing/<board>/`	Gerbers, separate plated/non-plated drill data, IPC netlist, schematic PDF, assembly PDF, all-parts and SMT/manual BOMs and placement files
`enclosure/keyboard-enclosure.FCStd`	Native FreeCAD full assembly, including all eight populated boards, screen, CM4, battery and speaker reference models
`enclosure/*.step`	Case, plate, spacers, straps, knob, light pipe, feet and custom heatsink geometry
`board-assemblies/`	Populated board STEP exports in each board's own coordinates
`bom/keyboard-parts-rev-h-audit.ods`	Parts, source addresses, listed prices, estimate ranges and manufacturer quote input cells
`bom/pcb-parts.csv`, `bom/assembly-parts.csv`	Procurement tables; exact MPNs or constrained procurement specifications
`assembly/`	Mechanical build, cable assembly, power constraints and prototype acceptance instructions
`firmware/`	RP2040/QMK/VIA and ESP32-C3 source, compiled images, programming instructions and source/version archives
`software/`	CM4 touch macro UI, local music integration, guarded target installer and deployment instructions
`checks/`,`CHECK-RESULTS.md`	Design-rule, connectivity, CAD, build and software-fixture evidence; physical tests are separately identified
`keyboard-layout.svg`, `key-positions.csv`	Key order, legends, widths and placement
`checks/audit-integrity.json`	Current native/source/export reconciliation; no release ZIP is approved

Open `kicad/keyboard/keyboard.kicad_pro` in KiCad 9; each daughterboard has its own project. Open the FreeCAD assembly for enclosure fit. STEP exports are interchange files, not a requirement to use a particular enclosure factory CAD system. All distances are millimetres. Revision H identifies this engineering audit. Some native board title blocks retain their

earlier individual board revision;`manufacturing/<board>/board-info.json` records it. The supplied file hashes identify the exact board versions reviewed, and earlier archive contents must not be mixed in.

Product requirements included in this revision

• 108 Cherry MX-compatible switch positions on standard **19.05 mm pitch**, with a 6.25u space bar. Cherry MX RGB Red clear-housing switches are the procurement baseline. Shine-through keycaps have the exact custom legends in the layout.

• F1-F12, main typing rows, Print Screen, 3 x 3 macro cluster, inverted-T arrows, full numpad and three dedicated Bluetooth selectors. Nine physical macro keys have four VIA-programmable banks.

• Per-key RGB through 108 addressable LEDs, with a firmware brightness cap and switched power. VIA configuration uses the wired PC connection; the keyboard also provides Bluetooth HID to three stored hosts.

• Five-inch portrait colour touchscreen, its top aligned with the F-key region, rotary control to its right, and a small battery light above the screen. Green = 80-100%; yellow = 30-79%; red = 0-29%. Invalid/stale battery telemetry turns the indicator off rather than showing a false charge state.

• Editable touchscreen macro pages, button colours, screen theme, brightness, IR learn/test/save, Wi-Fi device actions through Home Assistant, and a Music page for Plexamp.

• Internal Wi-Fi CM4 with 32 GB eMMC, so the screen and Plexamp run independently of the attached PC. The music library/Plex server must remain reachable, for example on a NAS; turning off the only library server also removes access to that library.

• Three USB-A downstream ports, a dedicated watch USB-C power output, a PC USB-C data input, a combined USB-C PD charging/data/programming input, and an OFF/ON switch beside the charging input.

• Two internal speakers, plus CM4 Bluetooth A2DP headphone output, for key sounds, local music and USB audio from the PC. PC audio needs the USB data connection. USB thumb drives enumerate on the attached PC; Bluetooth is not a wireless USB hub. The internal CM4 does not take ownership of those ports when the PC is off.

• Protected 2S rechargeable battery, monitored temperature, sequenced shutdown, and load shedding. Watch/USB/RGB/audio power availability depends on the power budget; see the power guide.

Remote-control scope: the installed hardware is infrared receive/transmit plus Wi-Fi integration. Point an existing IR remote at the keyboard to learn its command, or import a known supported code, then aim the keyboard at the appliance and test it. The user must confirm a response before saving. A passive projector cannot reveal its command simply by being pointed at. There is no RFID/NFC reader, universal RF scanner or automatic code discovery circuit in this revision. Wi-Fi bulbs require their supported Home Assistant integration and owner-supplied credentials.

Quote baseline

The spreadsheet's **approximately $745-$1,685 parts-and-case allowance** is a planning estimate, not a manufacturing quote. It excludes PCB fabrication, assembly labour, engineering/qualification, tax and shipping. Listed-price coverage is shown separately; blank catalog prices are not treated as zero-cost parts. Do not infer available stock from a catalog price. Supplier quotation and final purchased-MPN confirmation are part of procurement.

The manufacturer's handoff back to the owner should contain: the assembled prototype; updated native design and manufacturing files; purchased BOM; programmed firmware hashes; target OS image/version and commissioning instructions; completed measurements/test records; any deviations; and the priced path to a repeat build. Return it without the manufacturer's or another customer's credentials.

This project was developed and checked in an isolated workspace. No installation, flashing, service activation or operating-system changes were made to the owner's live PC.

Revision H open items - manufacturing HOLD

Updated 21 September 2026. An empty DRC report is not an electrical or product qualification. No replacement manufacturer ZIP is approved. Revision G must not be built. This list separates unfinished design review from measurements that need an actual prototype.

Design review before fabrication

Item	Current evidence	Required closure
Actual PCB stackup and signal integrity	Explicit native stackups; USB/DSI width corrections; connected-pair lengths checked; ideal cross-section estimates and filled-plane sampling recorded	Supplier must approve actual materials/pressed thickness/copper/etch and 90/100-ohm construction. Review the recorded connector/via reference interruptions, coupling changes, transitions and cables. Return any required native routing changes before fabrication. A local ideal estimate is not a qualified full-channel simulation.
Capacitors, ripple and regulator margins	AON regulator replaced by AP63203 2 A; current pin mappings checked; nominal capacitor banks identified	Obtain exact-part DC-bias/temperature/tolerance evidence and verify effective capacitance, ripple-current and load-step requirements. In particular, prove the three 4 x 47 uF TPS568230 output banks remain at least 66 uF effective. Input ripple, stability and transient review remains open. Nominal capacitance alone does not close it.
USB input startup and inrush	Diode-isolated hub supply, firmware charge qualification, PD sensing and current-limit fault fixtures	Review the complete input network, including USB-I/O C14/C30/C50, regulator startup and power-board C1/C2/C15 behind the PD FET. Establish cold-attach/default-current/suspend compliance and whether input limiting or a capacitance change is needed. Firmware HIZ policy does not limit every attach transient.
Battery and fault protection	Selected protected 2S CU-JAS503 pack, NTC interface, charger/gauge fixtures and load budget	Obtain the pack manufacturer's charge/protection/balancing/temperature/transport documentation and approve charger settings, fuses, connector ratings and fault interruption. The selected pack's dimensions and protection description are insufficient evidence.
Complete electrical design signoff	Eight clean native rule/parity/ERC checks; 100 selected IC pin checks; 12 hub strap/reset/bias resistor checks	Complete all remaining datasheet pin/rating, worst-case current, return-loop, tolerance, thermal and power-sequence review. Selected pin checks are not an audit of every component or an independent electrical signoff.
Purchased parts and enclosure process	829 positions reconciled; current 937 model references; nominal CAD fit checks	Confirm purchased MPNs and physical dimensions, tolerances, connector plug bodies, switch/stabilizer/keycap samples, FFC thickness/folds, materials, shrinkage and screw engagement. Simplified model envelopes are not supplier-certified metrology.

Prototype evidence still absent

All 26 rows in `assembly/acceptance-record.csv` remain NOT TESTED. There is no manufactured keyboard here. Required measurements include power/charge/thermal/fault performance; USB data and programming through both inputs; actual display and macro operation; Bluetooth keyboard plus headphones; speakers/audio; IR/Wi-Fi integration; runtime; and assembled mechanical fit.

These tests cannot truthfully be passed by adding more desktop fixtures. The firmware and software are built/tested against the documented fixtures only. The owner's live system has not been used as a test target.

Release gate

After the design issues above are closed, regenerate affected native checks, exports, BOM, models and test evidence. A specialist may then fabricate a controlled engineering prototype under the accepted bring-up plan. Production or an assertion that the entire product is fully tested requires the measured acceptance record and disposition of failures. A final release ZIP must contain only the mutually consistent approved revision, an inventory and verified hashes. No current archive is claimed to satisfy that gate.

Revision H audit: charging, programming and headphones

Manufacturing HOLD; physical operation remains unverified.

External USB connections

J6, beside the OFF/ON switch, is **CHARGE / DATA / PROGRAM**. It carries USB 2 data to the internal hub as well as USB-C PD charging. J1 remains an alternative **PC DATA** input. J5 is the **WATCH POWER** output for the owner's Samsung USB-C charging dongle. The three USB-A ports belong to the selected external computer.

J1 has automatic data priority whenever its VBUS is present. To program through J6, unplug J1. To use a dedicated 20 V / 3 A PD charger and a separate computer simultaneously, connect the charger to J6 and the computer to J1. The upstream switch selects one data host; it does not combine hosts or connect their VBUS rails directly. Changing the selected input disconnects/reconnects the hub, so safely eject storage first.

A cable must carry USB data for programming. A charger with no data host provides charging only. USB-C charging from a computer requires a source advertising at least 1.5 A or an adequate explicit PD contract. The firmware disables battery charging on an unqualified/default-current source. Ordinary low-power USB does not guarantee net charging while the CM4, display, music or peripheral charging are running. Use a 20 V / 3 A fixed-PDO source for the full power budget. Default-current/legacy USB use requires a charged, installed battery; startup and reconnect current still need prototype measurement before that source class can be accepted.

Programming through J6

1. Back up VIA settings and the CM4's private configuration. Eject USB storage. Unplug J1 and connect a verified data cable from the computer to J6. Use a qualified source and sufficient battery reserve.

2. **RP2040:** request QMK bootloader mode using a VIA-assigned Bootloader key. The supplied firmware requests an orderly CM4 shutdown and waits for its halt acknowledgment, with a 120-second fallback, before resetting. The independently powered hub stays supplied during RP2040 reset. Once the verified RP2040 ROM drive appears, copy `firmware/binaries/keyboard-rev-h-audit.uf2` to that drive. The native BOOTSEL/RESET buttons remain the recovery route; shut down the CM4 first because physical reset cannot wait for Linux. The SWD header remains available for initial bring-up or fault recovery.

3. **CM4 screen applications:** with the keyboard running, the same USB connection exposes a Linux serial console alongside stereo USB audio. Open that new serial device on the computer with a terminal supporting file transfer, and log in using the CM4 account/password. The target has `lrzsz` for ZMODEM upload/download (`rz` /`sz`); transfer application files, then restart the affected project service on the CM4. This is a normal authenticated Linux login. It does not expose a passwordless update shell or a network service to the computer.

4. **ESP32-C3:** upload replacement images and a reviewed matching manifest to the CM4. From its authenticated console, run `sudo python3 /opt/keyboard/rev-g/target/flash_esp32.py --bundle /path/to/verified/binaries`. For the shipped baseline, omit `--bundle`. The utility checks file hashes, exact offsets, controller telemetry and battery/PD reserve, stops the touchscreen service to release the serial link, uses the existing BOOT/RESET wiring, flashes the C3, resets it and restarts the service. Keep the OFF/ON switch on and power stable. The physical UART recovery header is still available if the C3 cannot communicate well enough for the guarded updater.

5. **CM4 eMMC recovery:** fitting carrier J9 and using the official `rpiboot` procedure exposes the CM4 storage through the hub. This is a service operation requiring the case to be opened; normal screen-code updates do not require it. Verify the target disk before imaging, then remove J9.

The packaged binaries were compiled and checked against fixtures. USB enumeration, recovery and programming on the assembled hardware are required acceptance tests, not completed physical measurements.

Two simultaneous Bluetooth roles

The ESP32-C3 provides keyboard BLE HID to the selected computer. The CM4's separate Bluetooth radio provides stereo A2DP audio **from the keyboard to headphones**. The headphones do not occupy one of the three keyboard host slots.

On the touchscreen, open Settings, put the headphones in pairing mode and choose **Find headphones**, **Pair and connect**, then select their audio output. Already paired headphones have a **Connect** button; **Disconnect** retains their bond, while

Forget removes it. Switching the output to **Keyboard speakers** returns playback to the internal speakers. Ordinary A2DP headphones are the target; headset microphone/call audio is not implemented.

Plexamp, key sounds and incoming PC USB audio share PipeWire. The Music tab remains available while macro pages and device controls are used. Plexamp can play with the PC off as long as the keyboard has power and its library server remains reachable. PC audio requires the USB data connection; the BLE keyboard link does not carry PC sound.

The target uses Raspberry Pi OS 64-bit Desktop **Trixie with WirePlumber 0.5 or later**, preserves CM4 Bluetooth, and assigns the ESP32 protocol to `/dev/serial0` with fixed core clock settings. Do not put a serial login console on that protocol UART. The login console above is `USBttyGS0`.

Hardware change and source basis

The USB board now uses TS3USB221A for data selection, AP63203 for a hub supply that survives keyboard reset, diode-isolated supply inputs and translated upstream-presence sensing. HUSB238 retains CC negotiation; its former DP/DM charging-detection pins are disconnected. The new parts and their models are included in the native KiCad project, BOM and populated board STEP. The enclosure openings and board mounting positions remain compatible.

The charger policy checks the HUSB238 status registers and measured VBUS before enabling charging. BQ25798 HIZ is reapplied for unqualified sources and ADC sampling restored after HIZ entry. Adapter attachment can reset HIZ, so firmware checks alone do not establish compliant startup current. Measure complete hub-plus-charger current, attach/inrush, suspend and weak-source behavior on the prototype. See the acceptance record for the required combinations.

Mechanical manufacture and assembly

Use the supplied native FreeCAD assembly and STEP solids. Logical layout X runs right and Y runs toward the front; in FreeCAD, Y is negative toward the front. Z is up. PCB placements are in `../layout-dimensions.json`; board STEP origins are local, while enclosure part STEP files retain assembly coordinates. Do not centre every imported file independently before assembling.

Material and dimensional baseline

Part	Quantity	Material / process	Requirement
Lower shell	1	Industrial SLS PA12, dyed black	Full 584.55 mm width in one piece; quote machine capacity. No unreviewed segmentation. Remove powder from passages and tap pilot holes after manufacture.
Top frame	1	SLS PA12, black	3 mm nominal cover; 2 mm remaining above rotary cable pocket.
Switch plate	1	1.5 mm 5052-H32 or 6061 aluminium, laser/waterjet/CNC	Deburr and flatten. Switch openings 14.00 mm, target +0.10/-0.00; validate the selected Cherry switch latch on a coupon.
Plate spacers	9	PA12 or machined insulating nylon	3.4 mm high, OD 5, bore 2.2; `PlateSpacer0` through`PlateSpacer8`.
CM4 spacers	4	Insulating nylon	3.0 mm high, OD 4.8, bore 2.7; do not confuse with 1.5 mm CM4 connector versions.
Speaker straps	2	PA12	2 mm thick; frame retention only, no cone load.
Rotary knob	1	PA12 or machined polymer	D-shaped bore; 20 mm OD, 12 mm high. Friction fit; do not drill away the flat.
Battery indicator light pipe	1	Clear PMMA or clear optical resin	Polish stem; diffuse visible 2.9 mm tip. Do not glue against LED packages.
CM4 heatsink	1	6061-T6 aluminium	`cm4-heatsink.step`: 24 x 24 x 6.5, 1.5 mm base, eight 1 mm fins. Deburr; base flatness 0.10. Add a 0.5 mm insulating thermal pad, total 7 mm.
Feet	4	Silicone rubber, adhesive backed	Diameter 12, height 4. Individual STEP references supplied; purchased equivalent permitted.

Case outline is 584.55 x 153.70 mm, with the display pod extending to 166.50 mm. Body plus cover reaches Z42; keys and knob project higher. Use ±0.30 mm general polymer prototype tolerance, but control PCB mounting centres to ±0.15 mm and mating heights to ±0.15 mm. The manufacturer must account for SLS shrinkage/warpage; these values are functional acceptance targets, not a claim that every printer meets them without compensation.

Plate thickness, support heights and PCB thickness are a stack: main PCB underside Z32.5, PCB top34.1, spacer3.4, plate underside37.5, top39.0. Changing any of these affects switch latching/travel. Main PCB is 1.6 mm. The 0.4 mm display-ribbon passage is especially sensitive: inspect the actual supplied FFC thickness, fold radius and latch engagement in a fit coupon before the full shell/cover build.

The custom heatsink fits inside the checked 24 x 24 x 7 allowance, including its pad. It does not establish cooling performance. Align its base to the CM4 processor lid, avoid adjacent parts, and verify interface contact without flexing the module. Preserve the underside ventilation openings and orient the fins to permit air circulation.

Hardware baseline

`hardware.csv` lists the fastening groups. Screws are stainless pan-head machine screws with small washers where stated. Thread the polymer pilots with controlled-depth taps after manufacture; do not install heat-set inserts into the existing pilot geometry. If the manufacturer prefers inserts for repeated servicing, return a revised boss design and fresh fit checks first.

Joint	Quantity	Baseline screw	Stack / engagement target
Cover to shell	See hardware.csv	M3 x 8	3 mm cover +0.5 mm washer; about 4.5 mm thread engagement
Main PCB + plate spacers + plate to posts	9	M2 x 10	6.5 mm hard stack; about 3.5 mm engagement; retain insulating PCB washers if screw head otherwise contacts copper
USB I/O PCB	5	M2.5 x 6	1.6 mm PCB; about 4.4 mm engagement
Power PCB	4	M2.5 x 5	1.6 mm PCB; about 3.4 mm engagement; no longer screw into bottom floor
CM4 carrier PCB	4	M2 x 4	1.6 mm PCB; about 2.4 mm engagement
Audio PCB	4	M2 x 4	1.6 mm PCB; about 2.4 mm engagement
Control PCB	4	M2 x 5	1.6 mm PCB; about 3.4 mm engagement
Rotary PCB	4	M2 x 5	1.6 mm PCB; about 3.4 mm engagement
Battery-light PCB	2	M2 x 5	1.6 mm PCB; about 3.4 mm engagement
Speaker straps	4	M2 x 5	2 mm strap, about 3 mm engagement; add compliant frame gasket without crushing speaker
CM4 module to carrier	4	M2.5 x 10, washer/nut	3 mm spacers, PCB/module stack; verify thread protrusion clear of enclosure
Display bosses from underside	4	M2.5 x 30 with washer adjustment	Through full-height hollow support; target 2-3 mm boss engagement, never bottom screw in display insert. Verify actual boss face/depth on the purchased screen.

Screw lengths above are nominal assembly selections. At the fit sample, check actual head diameter, washers, threaded insert depth and thread engagement with a depth gauge. Stop if a screw bottoms or a PCB bows. Set assembly torque by the selected polymer's thread pull-out test; there is no validated torque value yet. Do not use an arbitrary steel-joint torque on printed posts.

Assembly sequence

1. Inspect the shell for flatness, clear all passages, tap pilots to the measured safe depths, and trial-fit a blank PCB/plate coupon with the selected switches/stabilizers. Check the real USB plugs, ON/OFF actuator and display cable latch.

2. Fit keyboard stabilizers to the main PCB before installing its plate. Use seven 2u and one 6.25u Cherry-pattern PCB-mount stabilizer assemblies. Numpad Plus and Enter are vertical. Verify clearance through 4 mm travel; do not substitute different stabilizer hole patterns without revising the CAD.

3. Install power, USB I/O, audio, control, CM4 carrier, rotary and battery-light PCBs onto their labelled supports. The board-side copper/mounting-hole clearances are in KiCad; no screw may touch an exposed signal pad or trace.

4. Seat the CM4 into both 3 mm Hirose sockets with even pressure and no twist, fit its four spacers/fasteners, then fit the thermal pad/heatsink. The two connector-body overlaps in the CAD report are intentionally mated sockets, not an enclosure interference.

5. Mount speakers on thin soft frame gaskets and retain with the two straps. Leave the cones and vent openings free. Insulate both speaker leads: each amplifier output is bridge-tied; neither speaker terminal is ground.

6. Lay in the checked harnesses before the main keyboard PCB and display obstruct access. Follow `WIRING.md`. Fit the pack with soft foam/strap retention inside the 129 x 76 x 15 allowance; protect it from screw ends and sharp edges. Do not compress or puncture the pouch. Affix and insulate the NTC on the pack surface. Leave the battery disconnected until electrical bring-up authorizes it.

7. Fit display fasteners from the case underside, using washers to limit engagement. Use its supplied power lead by net assignment and its 22-to-15-way FFC. Route FFC beside the display backplate, then horizontally into the connector, following the CAD access allocations. Do not make the former direct vertical crossing through the backplate.

8. Fit the main PCB, nine 3.4 mm spacers and plate. Insert 108 switches squarely while supporting each hot-swap socket from behind; then fit the custom keycaps. Ordinary 1u caps are about 18.05 mm wide on 19.05 mm centres. Follow the actual key-width list, including both Shift keys and the 6.25u space bar.

9. Fit the light pipe above the indicator LEDs with an air gap; wires from this board exit toward the front. CM4 J10 wires exit toward the rear. Route rotary wires under the dedicated cover pocket. Fit cover and knob; require 0.8 mm knob press travel without cover contact and full key travel without rubbing.

10. Complete the acceptance checks with the enclosure closed. Fit four feet, record hardware/assembly deviations and return the final CAD if any dimensions changed.

The full FreeCAD assembly contains reference electronics and hidden cable allocation solids. Manufacture only the case/mechanical parts listed here. `keyboard-enclosure.step` is a combined mechanical reference, not an additional part to order on top of the individual STEP files.

Wire harness and connector assembly

The authoritative conductor map is `../harness.csv`, with a readable version in `../harness.md`. It contains 15 inter-board harnesses and64 conductors. `connector-pin-map.json` gives every native connector pad/net, including debug and external-module connectors. Numbers are the **KiCad pad numbers**, not an assumed left-to-right view of an unplugged housing. Verify both ends against the connector manufacturer's mating-face drawing and continuity-test each completed harness before applying power.

`cable-cut-list.csv` gives endpoint positions and prototype cut allowances. These lengths include service slack but are not a validated factory wire drawing: route one sample in the supplied case, trim/retain it without tension, and record the final finished lengths and dressing in the returned build record. Keep wires below the main PCB, away from switch tails, fanless cooling vents and display flex folds. Do not store excess wire over hot power components.

Housing/contact procurement

Board connector	Cable-side family	Conductors
JST BMxxB-GHS-TBT	GHR-xxV-S housing, SSHL-002T-P0.2 contact	26-30 AWG, insulation OD 0.76-1.0 mm; use 26 AWG for GH power and28 AWG signal. Maximum 1 A per contact with 26 AWG.
JST B2P-VH / B4P-VH	VHR-2N / VHR-4N, SVH-21T-P1.1 contacts	18 AWG copper for battery, core and IO power; current rating must cover the assigned circuit
2.54 mm single-row headers	Keyed/labeled crimp receptacle housing matching pin count and 0.64 mm square posts	28 AWG signal; 26 AWG when the other end is GH; appropriately rated24 AWG contact for audio supply. Add positive retention against accidental reversal.
Molex 53261-0271	Molex 51021-0200 housing and matching 50079-series PicoBlade crimp contacts	26-28 AWG paired speaker wire; verify exact contact plating/reel suffix during sourcing

H07 and H15 terminate into **one 7-position audio housing**: H07 fills pins1-2, H15 fills pins3-7. Do not buy two housings and try to overlap them. H04 uses a 3-way main-board housing; H05 an 8-way; H08 a 6-way. Every pad listed in a harness must be populated in its matching housing. Unused debug/RGB-chain headers remain accessible but have no routine harness fitted.

H04 and H14 carry USB differential pairs: use shielded 90-ohm differential USB cable with the D+/D- pair kept twisted, matched lengths and minimal untwisted connector fan-out. H14 also carries USB VBUS sensing. Connect shield/drain to the specified ground structure at the board entry; do not substitute shield for the ground conductor. GH/header transitions are not impedance-controlled connectors, so USB qualification is a required prototype test. Do not stretch the prescribed wire lengths to accommodate an arbitrary new case.

H15 I2S is a short digital harness. Keep it separate from inductor switch nodes, IR drive and class-D speaker pairs. Use its ground conductor adjacent to the clock/data group. Route both speaker leads together and neither to chassis ground.

External module leads, additional to H01-H15

• Battery: power-board J8, pad1 positive and pad2 GND **only after checking native pin map**. Use the selected pack's existing protection circuit; no bare unprotected cell assembly. Connect a matching VH housing and strain relief without solder heat reaching the pouch.

• NTC: separate 2-wire lead to power-board J9, for the cell-mounted103AT-2. It is not the battery's middle-cell tap. Continuity/temperature-test independently; an open/invalid NTC must inhibit charging.

• Display: CM4-carrier J5 pin 1 is +5V_DISPLAY, pin 2 unconnected, pin 3 GND. Place the red lead on pin 1 and black on pin 3 by their nets; the display's supplied housing does not imply correct orientation on this custom3-pin header. Fit the supplied 22-to-15-way DSI cable; the 22-pin end goes to carrier J8 and 15-pin end to the display. Carrier J9 is the eMMC recovery jumper, not the display connector.

• Speakers: two PicoBlade2-wire leads from audio J1/J2 to the selected PUI drivers. Keep left/right and phase consistent; both terminals are active amplifier outputs.

• CM4 antenna and ESP32 antenna: separate RF pigtails/antennas. Use the actual module-approved antenna types, place them against polymer and clear the battery/plate. Include their installation in the radio acceptance measurement.

All interconnect grounds in the pin map are required. Never join rails because two connectors use the same housing. Never connect a 5 V USB-UART adapter's supply pin to 3.3 V debug power. Programming adapters may share ground and 3.3 V logic; leave their supply pin disconnected when the board is already powered.

Label external rear ports permanently: **USB-A 1 / USB-A 2 / USB-A 3 / WATCH5V / PC DATA / CHARGE PD / OFF-ON**. The watch port is an output; the charging port is the PD input. These labels must remain visible after assembly.

Electrical fabrication and power baseline - Revision H audit

Manufacturing HOLD. The following requirements do not authorize a build. See `../OPEN-ITEMS.md`.

Board fabrication

Manufacture one of each of the eight board designs. The seven non-audio boards have four copper layers; stereo-audio has two. All are nominal 1.6 mm FR-4. Copper layer order is F.Cu / In1.Cu / In2.Cu / B.Cu for four-layer boards, F.Cu / B.Cu for audio. Match each board's Gerber job and native stack definition. Use ENIG, matching the current native stackup. See `../manufacturing/STACKUP-REQUIREMENTS.md` for the explicit laminate and controlled-impedance requirements; supplier approval remains open. Quote 1 oz nominal external copper and the native internal copper requirement, solder mask both sides and white legend. No blind/buried vias are specified. Plated and non-plated drill files are separate; preserve routed slots and mounting holes.

Board outlines are the Edge.Cuts **centre lines**. Gerbers and drills use the absolute native board origin. Placement files use millimetres, positive X right, positive Y up as viewed from the front; bottom rotations are native KiCad, not mirrored. The assembly house must verify its library zero-angle conventions in its placement preview, especially hot-swap sockets, reverse-mounted RGB LEDs, USB connectors, QFNs and the 200-pin CM4 connector pair.

Use `all-bom.csv`/`all-cpl.csv` as the complete 829-position inventory across the eight boards. `jlc-smt-*` excludes footprints with plated through-hole pads;`manual-*` contains those mixed or through-hole assemblies for manual/selective solder. A connector with both SMT and through-hole anchors appears in the manual group so it is not silently omitted. The manufacturer may move such parts into its qualified pin-in-paste process, preserving counts. Mounting holes have no purchased component and are absent from BOM/CPL. There are no intentionally omitted electronic parts in this release.

`bom/pcb-parts.csv` is the procurement authority for assigned MPNs/specifications, including updated passive selections. Schematic values describe the circuit; a higher-voltage capacitor in the same package is intentional where the procurement table selects it. Parts whose MPN starts with `PROCUREMENT SPEC` must be sourced by the manufacturer to that complete specification and returned with the purchased MPN/LCSC code. Do not fabricate catalog IDs. The quote-stage assignment does not authorize a different IC pinout, capacitance, resistor value, connector mating height or fuse rating.

The copper uses regular horizontal/front and vertical/back routing for ordinary signals. Compact pad escapes, power distribution, ground planes and high-speed pairs are exceptions. **Native DRC does not establish differential impedance or high-current thermal performance.** Before releasing Gerbers to fabrication, the manufacturer's electronics engineer must approve a real fabrication stackup, compare routing to 90-ohm USB /100-ohm DSI differential targets, review reference-plane continuity/via transitions and QFN thermal/return paths, and return any required corrections with new DRC/ERC and enclosure results. Do not change trace widths or internal layers silently during CAM processing. Controlled-impedance coupons and the prototype link tests are the acceptance evidence.

Bare boards require electrical open/short testing using the supplied IPC-D-356 netlist. Assembled boards require AOI, polarity review, continuity and X-ray of hidden solder joints where optical inspection cannot establish wetting/bridging. Respect package MSL/bake/reflow and LED/switch-socket heat limits. No glue or conformal coating on hot-swap contacts, USB mating contacts, FFC latches or RF sockets.

Rails and component constraints

Rail/function	Baseline
Input	Dedicated USB-C PD, validated 20 V /3 A offer required for full input-power mode; HUSB238_002DD negotiation and independent voltage telemetry
Pack	Protected 2S, 7.4 V nominal /8.4 V charge ceiling, 5 Ah CU-JAS503, 6 A maximum specified discharge
Charger	BQ25798, 2S configuration/readback, 1 A charge request; temperature/fault gating and watchdog
Always-on	AP63203WU-7, 3.3 V; 2 A regulator class; verify ESP32/radio/RP2040 transient headroom before connecting all loads
Core and IO	TPS568230 regulators, 5.1 V nominal feedback; output banks 4 x 47 uF each
Audio	TPS568230, 3.3 V nominal, 4 x 47 uF bank; two MAX98357A amps into 8-ohm speakers
RGB	TPS2553 limited switched rail, brightness cap 32/255 in QMK; 108 SK6812MINI-E LEDs
External USB	Three USB-A outputs, budget 1.5 A combined; per-port limiting/fault signals
Watch	TPS25810 USB-C source, 5 V /1.5 A class, switched by the supervisor; no watch dongle protocol is implemented in keyboard firmware

Regulator feedback uses the selected 0.1% resistors. For 5.1 V, 75k/10k with 0.6 V reference gives 5.1 V nominal; for 3.3 V, 45k/10k gives 3.3 V nominal. These are nominal calculations, not measured tolerances. The 4 x 47 uF bank is 188 uF nominal; the selected capacitor's DC-bias, temperature and tolerance must leave **at least 66 uF effective per buck bank**, following the TPS568230 recommended output-capacitance range. Do not replace with physically smaller high-bias-loss parts based on "47 uF" alone. Check the 1.0/1.5/2.2/4.7 uH Bourns inductors against actual ripple, peak/saturation, RMS and temperature rise; their schematic current labels are requirements, not proof of assembled-circuit current capability.

The input fuse is 3 A; battery fuse 10 A is for fault interruption, **not permission to draw 10 A continuously from a 6 A pack**. The firmware trips on sustained overcurrent and budgets below the pack's stated limit. Validate fuse interruption behaviour, MOSFET SOA, connector/trace/via heating and charger thermal dissipation with the real stackup and enclosure.

Supervisor behaviour implemented in the supplied firmware

• Validate charger identity/cell configuration, battery voltage, NTC and status before enabling charge or loads. Defaults are disabled; failed readback or lost control bus disables discretionary outputs.

• Initialize the charger input limit at 100 mA. Unqualified/default-current input stays in HIZ with charging disabled. A confirmed USB-C source of at least 1.5 A, plus measured VBUS matching its contract, permits a 500 mA charger input limit. Only an accepted 20 V / 3 A result and measured 18-21 V input permits 2.3 A. The service hub draws independently; measure combined input current, inrush and HIZ reset-on-attach behavior before accepting any host/source class.

• Charge voltage is 8.4 V and request 1 A. A valid temperature/fault state is required; do not bypass NTC checks to make a prototype appear to charge.

• Core start from battery requires at least 6.4 V and more than 8% SOC; shutdown is requested at low pack voltage/SOC or faults. The CM4 acknowledges the current boot generation, then performs an orderly shutdown; the supervisor waits for the final halt handshake before removing the main rails.

• The high-PD budget is 36 W. Reserve 17 W for core/display/hub, then grant audio 3 W, RGB 3.5 W, external USB 7.65 W and watch 7.65 W in priority order. All loads together exceed this reservation, so **the watch can be shed when all other loads are requested**, even on PD. The screen distinguishes requested and granted power.

• Battery budget uses pack voltage x 5 A x 0.85 minus 0.8 W before reservations. This is a conservative allocation model, not a runtime guarantee. Do not raise it without measurements and engineering review. At lower charge levels, disconnect external charging loads before sacrificing orderly shutdown.

Manufacturer engineering evidence required

Record rail min/max/ripple and brownout margin, input/pack current, inductor and connector temperatures, effective-capacitance evidence, charge behaviour with the selected pack, fault response, and shutdown timing. Validate the battery supplier's protection and charging documentation, including whether balancing is present/required. The selected pack's bounding dimensions and protection claim alone do not close that electrical review. Test the USB-C watch port with the owner's Samsung charging dongle; the exact watch/dongle SKU was not supplied.

Primary references: BQ25798, TPS568230, AP63203, TPS2553, TPS25810, Bourns SRP7050TA, CM4 hardware. The native schematics and firmware define this design's actual pin assignments and limits.

Combined USB input

See USB-C and headphones for input selection and programming. Verify AP63203 3.3 V hub regulation at minimum input through SS14, diode heating, SRP5030T-4R7M ripple current, and the effective capacitance/transient response of its 2 x 22 uF output bank.

Programming and firmware sources - Rev H audit

Manufacturing release on hold.

The supplied RP2040/QMK and ESP32-C3 firmware **compiled successfully**. Their source, binary hashes and upstream versions are included. They have not been flashed or validated on a physical prototype; perform the bring-up/acceptance plan before connecting the real battery or peripherals.

`binaries/manifest.json` is the image authority. RP2040 uses the 16 MiB W25Q128JVSIQ (128 Mbit, not 128 MB). ESP32-C3-WROOM-02U-N4 has 4 MiB flash. The internal CM4 uses its 32 GB eMMC; no microSD slot is fitted.

RP2040 keyboard

Image: `binaries/keyboard-rev-h-audit.uf2`. QMK target`personal/full_keyboard` , keymap`via`. The image includes matrix/RGB, four macro banks, VIA dynamic configuration, Bluetooth bridge, guarded macro taps, power supervision and shutdown sequencing.

Use a suitable 3.3 V SWD programmer on keyboard J1 for first controlled bring-up, or use BOOTSEL/RESET to enter the RP2040 ROM UF2 USB mode after the board has a qualified supply and the internal hub/USB connection is working. J1 pin 1=3.3 V reference, 2=GND, 3=SWDIO, 4=SWCLK, 5=RUN, 6=GND. Do not inject a second supply into an already powered board. For UF2, hold BOOTSEL while resetting, release when the RP2040 ROM drive appears, and copy the image to that **verified target drive only**. Do not write a raw firmware image to an arbitrary PC disk.

To rebuild in a separate development environment, install the QMK build prerequisites for your OS, extract `upstream/qmk-build-sources.tar.gz`, and build with `qmk compile -kb personal/full_keyboard -km via` from the extracted QMK tree. The pinned commit/submodule revisions are in the binary manifest. The `qmk/` folder is the standalone project source for edits. Copy its contents into`keyboards/personal/full_keyboard` when starting from a clean upstream checkout. Preserve the pinned revision and required ChibiOS/PicoSDK submodules.

VIA definition: `qmk/via-definition.json`. Connect through J1 PC DATA, or J6 CHARGE / DATA / PROGRAM with J1 unplugged, use VIA's design/import facility for this custom definition, then configure keys/banks/RGB. VIA over Bluetooth is not implemented. PrototypeVID/PID values must be replaced with a legitimate allocation before commercial distribution; update the VIA definition consistently.

BT1-BT3 short press selects a stored Bluetooth host; hold about 1.8 s to pair that slot. Fn+BT1 selects wired USB. Rotate the encoder in macro mode to select one of four macro banks; holding its button about 1 s changes between macro and touchscreen-selection modes. The 9 physical macro keys are read from the chosen VIA bank; default banks send F13-F21 with different modifiers.

ESP32-C3 control/radio board

The build is pinned by `esp32/platformio.ini`: Espressif32 platform 6.12.0, Arduino framework 2.0.17 package revision, NimBLE-Arduino 2.5.0, ESP32-C3 target, DIO flash mode, 80 MHz flash, 4 MB flash. Rebuild with `pio run -d esp32` in a separate build environment. The module uses an external antenna; do not substitute a non-U module footprint.

Factory programming uses control-board J6 with a 3.3 V UART adapter: pin 2 GND; adapter RX receives pin 3 C3_TX; adapter TX drives pin 4 C3_RX. Pin 5 is EN and pin 6 BOOT_N. Pin 1 is 3.3 V reference/supply; leave the adapter's supply disconnected when the board is powered. Disconnect the CM4 serial harness or keep its TX inactive during this procedure. Hold BOOT low, pulse EN low then high, release BOOT after entering ROM download mode.

Use the selected adapter port explicitly. Example command, run **only with the prototype connected and the port verified** from `binaries/`:

```
python -m esptool --chip esp32c3 --port TARGET_UART_PORT --baud 115200 write_flash --flash_mode dio
--flash_freq 80m --flash_size 4MB 0x0000 esp32/bootloader.bin 0x8000 esp32/partitions.bin 0xe000
esp32/boot_app0.bin 0x10000 esp32/firmware.bin
```

The offsets come from this exact PlatformIO build's flash images/default partition map, recorded in the manifest. Reset EN after successful write/verify. Do not use the original ESP32's 0x1000 bootloader offset on an ESP32-C3. Reflashing the complete partition set may reset pairing/configuration; preserve owner data for upgrades and test the update process before deployment.

The ESP32 implements authenticated three-slot BLE HID, packet forwarding, rotary reading, battery-colour PWM and 38 kHz IR envelope learning/transmit. Raw imported IR carriers are bounded; the 38 kHz receiver does not measure arbitrary carrier frequency or capture RFID.

CM4 and software

For updates through the existing charging port, see USB-C and headphones. The software bootloader path now waits for CM4 shutdown before resetting.

Provision the CM4 using `../software/README.md`. The keypad and radio microcontrollers remain independently programmable, while the CM4 screen is a normal Linux software environment for future applications. The existing touch UI is source code, not a fixed Plexamp-only appliance.

Build/relink sources and licensing

Custom QMK/ESP32 files carrying SPDX GPL-2.0-or-later retain that licence. The bundled QMK build source archive includes the target, core/build files and used ChibiOS/PicoSDK/printf source trees with their existing licences. The ESP32 source archive includes NimBLE, Arduino core/libraries/variants and relinkable project objects. PlatformIO downloads the pinned compiler/framework SDK; its upstream package source/version is in the manifest. Keep all third-party licence notices when redistributing binaries or derivative sources. Obtain/update the matching upstream SDK source as its licence requires when shipping a modified binary; do not treat the keyboard's product files as a transfer of third-party rights.

Plexamp and Node archives are **not redistributed** here; the target installer fetches their pinned official releases and verifies SHA256 from `../software/release-manifest.json`. Account credentials, Wi-Fi passwords and Home Assistant tokens are never supplied in this archive.

Internal computer and touchscreen deployment

The CM4 runs the touchscreen application and Plexamp independently of the attached PC. The touch application includes programmable macro pages and device controls; the Music tab does not replace those functions. Wi-Fi connects this internal Linux computer to the owner's network. The ESP32 handles the keyboard's separate Bluetooth host link.

The application and installer were checked with isolated simulation/fixtures. Audio hardware, device-tree overlays, radio operation and physical shutdown have not been exercised on a built keyboard. The manufacturer must complete `../assembly/ACCEPTANCE.md` on the final board stack.

Hardware-free preview

From this folder, run in an isolated development directory:

```
python3 -B -m keyboard_ui.app --simulate --data ./preview-data
```

Open `http://127.0.0.1:8765` in a browser. Simulation does not open serial/GPIO or send Wi-Fi/IR commands. It saves only the chosen preview data. Stop the process when finished. The Music tab requires the separately installed Plexamp service; simulation does not install it.

To inspect exactly what target deployment files would be prepared:

```
python3 -B install_target.py --prepare-directory ./fresh-target-preview --desktop-user keyboard
```

The destination must be empty or new and must not resolve through symlinks. This command creates files inside that directory only; it does not download releases, install packages, enable services, change boot settings or flash a device.

Provision the dedicated CM4

These steps belong on the **new keyboard CM4**, not on the owner's live computer. Use a current supported **64-bit Raspberry Pi OS Desktop Trixie, with WirePlumber 0.5 or later** image with KMS and the five-inch Touch Display 2 overlay. Record the exact image release/kernel/hash in the manufacturer acceptance record. The installer expects `/boot/firmware` and Python with `tarfile.data_filter` support (Python 3.12 or a maintained 3.11 version with that backport).

1. Power the carrier using the qualified prototype supply arrangement. Fit the carrier J9 recovery jumper only for eMMC programming and reset the CM4 into USB boot. Use the official Raspberry Pi `rpiboot` procedure and verify the target storage identity before writing the 32 GB eMMC. Remove J9 afterward. The carrier's internal USB link reaches the PC through the hub; if initial hub power is not yet qualified, use the carrier's documented USB debug connection under the hardware engineer's control. Do not connect two USB hosts simultaneously.

2. Create a normal desktop user, for example `keyboard`, with a private password and appropriate country/timezone/Wi-Fi settings. No credentials are included in this archive. Configure networking on the CM4 through Raspberry Pi OS. Use the KMS desktop configuration; FKMS is rejected by the installer.

3. Copy the package to the CM4, retaining `software/` beside `firmware/binaries/`, then enter `software/`. After the manufacturer has reviewed the target changes, run there, substituting the actual regular desktop username:

```
sudo python3 install_target.py --apply-on-cm4 --desktop-user keyboard
```

The script checks the CM4 model, ARM64 architecture, root permission, regular user home and boot configuration before package installation. It installs Python, PipeWire/WirePlumber, BlueZ, ALSA compatibility, USB file-transfer and Chromium dependencies, downloads pinned official Node 20/Plexamp releases with SHA256 verification, prepares project files under `/opt/keyboard`, backs up the affected initial boot/audio/desktop configuration under `/var/backups/keyboard-revg`, writes project services, and adds the user to audio/video/serial input groups. It enables the dedicated user's PipeWire session and Bluetooth audio, and enables the USB serial login console. This installer belongs only on the dedicated keyboard CM4. Services start after the next target reboot; Plexamp is left disabled until claimed.

4. Enable desktop autologin for this dedicated appliance using Raspberry Pi OS's configuration utility. Complete the Plexamp claim as the desktop user, following the official headless setup flow. From `/opt/keyboard/plexamp`, run `/opt/keyboard/node/bin/node js/index.js` interactively for initial setup; keep claim tokens out of logs and the handoff archive. Select the intended Plex library/server and the ALSA default output routed through PipeWire. Stop that interactive process after setup, then enable its project service:

```
sudo systemctl enable keyboard-plexamp.service
```

5. Reboot the CM4. Verify the local touch page at `http://127.0.0.1:8765` and the Plexamp player at`http://127.0.0.1:32500`. Confirm that only one Plexamp process runs. The owner needs a Plex account and a

reachable library/server; any account/subscription requirements are governed by the current Plex product, not granted by this hardware package.

Display, serial and audio configuration

`target/boot-config.txt` defines the five-inch DSI display, MAX98357A I2S, USB device controller, UART and final-halt GPIO. CM4 Bluetooth remains enabled for headphones. The ESP32 protocol uses the mini UART through `/dev/serial0`, with fixed core clock settings; no Linux login console may use that protocol UART.

The display connects to DSI1 at carrier J8. Carrier J5 supplies display power on pin 1 and ground on pin 3. The installer adds a separate boot include and rejects conflicting FKMS or explicit Bluetooth-disable settings.

The USB gadget exposes stereo UAC2 audio (48 kHz, 16-bit) and an authenticated serial console. It binds only while upstream VBUS is sensed. `keyboard-audio-loop.service` finds the UAC2 capture node using PipeWire device metadata and routes it to the selected default output. Plexamp and key sounds use that same PipeWire session. The target can select speakers or Bluetooth A2DP headphones in Settings. Confirm actual source/sink enumeration, clock adaptation, no-underrun mixing and radio coexistence on the assembled hardware.

USB-C and headphones contains the port-selection rules, power requirements, update procedure and headphone controls. Both radios can operate independently: ESP32 BLE HID to the PC and CM4 stereo audio to headphones. USB PC audio is separate from BLE HID.

Owner configuration and future coding

• **Macros:** use the touch editor to create pages/buttons, colour them and add shortcut, text, delay, bank, saved IR or Home Assistant steps. Export the macro configuration before updates. Text macros currently use printable ASCII with the US keyboard layout; OS layout changes require matching mapping work.

• **Device controls:** configure the owner's Home Assistant base URL and long-lived access token locally. Discover supported entity names through that authorized integration, then assign actions. Bulb brand/app credentials are supplied by the owner inside Home Assistant; arbitrary Wi-Fi devices are not automatically controllable just because they appear on a network.

• **IR:** learn from an existing 38 kHz remote or import a validated Pronto/NEC/raw code. Test while aiming the keyboard at the device; the save action requires the owner's explicit successful-response confirmation. No automatic visual response detector is installed.

• **Power:** the screen can request audio/RGB/USB/watch loads and shows the supervisor's granted state. Disable higher-priority discretionary loads if the watch output is shed. The OFF/ON switch requests a safe shutdown; charging may remain possible while the main keyboard is off.

• **Future applications:** edit `keyboard_ui/` for screen behaviour,`firmware/qmk/` for matrix/VIA/power policy and`firmware/esp32/` for BLE/IR/encoder logic. Back up persistent configuration and use the hardware-free tests before a controlled target update. Do not raise power/charge limits casually to support a new application.

Persistent target macro/IR/integration data is in `/var/lib/keyboard`; it contains private tokens when integrations are configured. The UI binds to localhost and checks mutation tokens; it is not an internet-facing control service. Backups of owner data must remain private and should not be included in manufacturer source archives. Linux OS/Plexamp credentials and pairing state are provisioned per unit.

Failure diagnosis / return to baseline

Inspect the target journals for `keyboard-ui`, `keyboard-usb-audio`, `keyboard-audio-loop` and `keyboard-plexamp`. Check UART, power grants, display overlay and audio enumeration first. Keep credentials out of reports. Restore failed deployment from the target image or `/var/backups/keyboard-revg`; never run the installer on another PC. Return the commissioned image and private credentials securely.

Official references: CM4/eMMC USB boot, Touch Display 2 setup, Plexamp. Exact downloaded Node/Plexamp versions and SHA256 values are in `release-manifest.json`.

Prototype acceptance and manufacturing release record

Status at handoff: physical tests have not been performed. Native electrical/CAD checks and software fixtures are documented separately. The specialist manufacturer owns the physical build, measurement and integration work below. Use `acceptance-record.csv` to record serial number, measured results, instrument/setup, operator, date and evidence path. Marking a row "pass" requires its evidence; "not tested" is not a pass.

A - before making or powering hardware

1. Reconcile the purchased BOM against 829 positions, all 8 board exports, their hashes and the manual/SMT split. Check exact package pin numbering against datasheets for charger/regulators/gauge/PD/hub/level translators/CM4 connectors. Approve all PROCUREMENT SPEC MPN assignments and any stock alternatives; retain the original design value and rated voltage/dielectric/current requirements.

2. Approve actual layer stackup, copper thickness, return planes and 90-ohm USB /100-ohm DSI routing. Assess decoupling loops, switch-node coupling, current paths, QFN exposed-pad vias, copper and connector heating. Record any routing correction as a new revision and rerun DRC/ERC, harness and mechanical checks. Do not use a zero DRC count as an electrical-rating signoff.

3. Obtain pack charge/protection/temperature and transport documentation. Confirm the 2S 8.4 V/1 A charge profile and 6 A discharge limit are supported, clarify cell balancing, and verify thermistor mounting/curve. Verify fuse coordination and USB-C fault behaviour. Do not connect the real pack to an unqualified charger.

4. Build small fit samples covering switch/stabilizer/plate, USB openings, display mount and FFC corridor, rotary pocket, indicator light pipe, screw engagement and CM4 spacing. Verify connectors with actual mating housings and purchased part tolerances, not only the simplified 3D envelopes. Approve case material/process shrinkage and screw torque after coupon tests.

B - bare boards, assembly and controlled bring-up

5. Bare-board open/short test against IPC-D-356. Verify fabrication dimensions, plated slots and mounting holes. Assembled-board AOI/X-ray as appropriate; prove no power-to-ground shorts or reversed electrolytic/LED/diode/IC orientations. Record resistance readings before power; capacitor charging means there is no universal resistance threshold for every rail.

6. Start with the real battery and external loads disconnected, using a current-limited battery simulator/bench arrangement under the electronics engineer's control. Verify 3.3 V always-on supply before connecting programming tools. Program RP2040 and ESP32 by the documented debug/ROM paths; never use an uncontrolled live USB supply to backfeed the rails. Confirm regulator defaults remain off until supervisor checks pass.

7. Measure3.3 V AON, 5.1 V CORE, 5.1 V IO, 3.3 V AUDIO and RGB rails at source and far load. Accept within each receiving component's datasheet limits under idle, startup, key/RGB/radio bursts and maximum granted loads; also record ripple/transient droop. Establish current limits from the schematic/device ratings before raising bench limits. Test AON radio burst headroom specifically: its revised 2 A class regulator is not an assumed infinite source.

8. Verify 5 V USB input fallback and 20 V PD offer detection independently from telemetry. Reject false/missing PD results without requesting excessive input current. With a suitable simulator, check charger voltage/current regulation, termination/recharge, open/hot/cold NTC, I2C timeout, watchdog reset, unplug/replug and pack protection behaviour. Only then repeat qualified tests with the selected protected pack and attached thermistor.

9. Test short/overload behaviour separately on every USB-A and watch output using a protected load fixture. Confirm no backfeed into the PC or PD input. Measure combinedUSB-A current limit and watch CC source advertisement. Attach the real Samsung USB-C charging dongle and confirm the watch charges without resets or overheating.

C - user functions

10. Scan all 108 keys against the layout, including both Shift keys, numpad0/Plus/Enter, Print Screen, Bluetooth selectors and all 36 macro-bank assignments. Validate ghosting/diode orientation, rollover, host LEDs, held-key host switching and release behaviour. Program and read back a VIA change after a power cycle using the wired PC connection.

11. Illuminate each RGB LED individually, then use capped all-white and animation patterns. Verify ordering, colour channels, no flicker at radio/load bursts, regulator/connector temperatures and disabled-rail behaviour. Do not defeat the brightness cap for this test.

12. Pair three distinct hosts, switch between all slots repeatedly, reconnect after sleep/power loss, repeat passkey pairing, and verify reports reach only the selected authorized host. Test wireless range in the closed case with both final antennas installed. Confirm USB host mode via Fn+BT1 or the touch selector.

13. Exercise touchscreen corners, on-screen typing, macro editing/import/export, 12 pages/24 buttons bounds, shortcut/text/delay actions, colours/brightness, rotary macro-bank mode and screen-selection mode. Long press around 1 s changes mode; short press activates a selected on-screen control. Ensure a cancelled/host-switched queued macro does not execute later and held physical modifiers do not leak into macros.

14. Inject/report SOC values 100, 80, 79, 30, 29, 0 and invalid/stale telemetry. Require green at 80-100, mixed yellow at 30-79, red below 30 and off for invalid/stale data. Check visibility/mixing through the actual tiny light pipe at ordinary viewing angles. Calibrate gauge behaviour against actual pack discharge; boundary logic alone does not establish SOC accuracy.

15. With the PC USB data cable, confirm all three thumb drives/peripherals enumerate and sustain simultaneous read/write without errors. Verify repeated insertion/removal/suspend/resume, qualified cables and hubs. Confirm a failing/overcurrent device cannot reset the whole keyboard. USB2 high-speed testing, endpoint descriptors and signal quality require actual hardware evidence.

16. Play test left/right tones, keyboard clicks, PC USB stereo audio and local Plexamp, separately and mixed. Confirm channel order, sample rate 48 kHz stereo, acceptable latency/no underruns, no clipping at the intended maximum volume, and muted startup/shutdown. Check speaker clearance and rattles with the case closed. Confirm CPU/USB audio clock adaptation over a long run.

17. With the PC switched off, use the CM4 touchscreen and Plexamp against a still-running NAS/Plex server. Record network/library prerequisites and account claim. Test UI access to macros while music runs. The keyboard is not a Plex server and cannot access a powered-off library server.

18. Learn a known 38 kHz IR command from its existing remote, transmit toward the intended device, and save only after confirming success. Test timeout/malformed/imported-code limits. Configure a test Home Assistant integration for a Wi-Fi bulb and verify selected entity on/off/toggle. Preserve owner opt-in and never bundle credentials. Do not advertise universal RFID/RF learning.

D - closed-case power, thermal and recovery

19. Run worst granted loads with charging, RGB, wireless, display, audio and USB peripherals. Record all input/pack currents, rail transients, heatsink/inductor/regulator/connector/pack/case temperatures until stable. Pass only within component and battery supplier limits, without throttle-induced malfunction, audible instability, brownouts or unsafe touch temperatures. Ambient, orientation and permitted airflow must be recorded. If cooling is insufficient, revise the enclosure/thermal design and recheck before closing this item.

20. Exercise all combinations of requested loads. Verify the 36 W PD reservation, battery budget, watch-first shedding when the other reservations exhaust headroom, fault messages and recovery. Record measured battery runtime under at least typing-only and display/music use; do not promise runtime from capacity arithmetic alone.

21. Exercise switch-off, low battery, interrupted communications, stale boot-generation acknowledgements, CM4 boot failure and forced emergency power cut. Require orderly filesystem shutdown where energy/time permits, no reboot loop at low charge, and explicit OFF/ON rearm after a latched low-battery stop. Inspect storage after repeated cycles.

22. Demonstrate RP2040 recovery, ESP32 ROM reflash and CM4 eMMC reimaging with the actual assembled access points. Verify all programmed-image hashes and factory-default configuration. Assign unique product identifiers/serials and use licensed USB VID/PID before commercial distribution; the included prototype IDs are not an allocation.

23. Finish a full closed-case fit/travel/rattle check, cable retention/pull test and appropriately specified transport/drop tests for the intended product. Decide applicable EMC/radio/electrical/battery compliance requirements for the actual sales market before any volume build.

Return evidence and disposition

The prototype may be accepted only after every required row is passed or a specific owner-approved deviation is documented. Return measured results, photos, firmware/OS versions, purchased BOM, final cable lengths, torque limits, production notes and corrected native files. A finding requiring a component, copper, firmware or case change triggers focused retesting of the affected interfaces and a fresh final rule/fit/package audit.

E - USB and headphone acceptance

24. USB inputs: qualify J6 alone, J1 alone, and both connected with J1 priority; no backfeed or host shorting. Measure complete input current at cold attach, reset, enumeration, suspend and hot replug on default USB, 1.5 A Type-C and 20 V / 3 A PD. Confirm HIZ/charge policy and hub 3.3 V stability; record supported source classes.

25. Programming: through J6 with J1 unplugged, enter RP2040 ROM mode after orderly CM4 halt and reflash the UF2. Demonstrate authenticated USB serial file transfer, guarded ESP32 update and physical recovery fallback. Verify hub continuity through reset and intact CM4 storage after repeated cycles.

26. Headphones: pair, disconnect, reconnect, forget and switch outputs on actual A2DP headphones. Play Plexamp with PC off, then mix USB PC audio and key sounds while ESP32 BLE HID remains connected. Verify closed-case range, channel order, latency, no underruns and UART reliability at the fixed clock.

Verification results - Revision H audit

Updated 21 September 2026. **Manufacturing HOLD.** Current native files, exports, nominal CAD and fixture evidence are reconciled below. No physical keyboard has been built or tested, and no replacement supplier ZIP is approved. Open electrical design work is listed in `OPEN-ITEMS.md`.

Check	Result and scope	Evidence
Native KiCad 9 checks	All eight boards: zero DRC violations, unrouted connections, schematic-parity mismatches and ERC violations	`checks/current-h-native/summary.json` and per-board reports
Strict footprint/courtyard checks	Zero errors/warnings with the normally ignored footprint/courtyard checks enabled in isolated copies; no DRC/ERC exclusions	`checks/strict-rule-audit.json`, `checks/model-and-rule-audit.json`
3D model references	937 references resolve; no populated component lacks a model	`checks/model-and-rule-audit.json`
Harnesses	15 harnesses / 64 conductors match the current native connector pins	`checks/harness-check.json`
BOM and placement	829 fitted positions; complete/SMT/manual placement and BOM groups agree	`checks/bom-cpl-audit.json`
Stackup and pair routing	Explicit layer constructions; USB/DSI widths corrected; connected paths and CM4 package-adjusted pair lengths pass the stated limits	`checks/signal-paths.json`, `manufacturing/STACKUP-REQUIREMENTS.md`
Impedance/reference screening	Ideal cross-section estimate has numerical convergence checks; filled-plane sampling identifies remaining launch/transition review areas	`checks/impedance-estimates.json`, `checks/reference-plane-sampling.json`. Not a physical SI pass
Selected IC pin review	100 pad-to-net checks plus 12 USB hub strap/reset/bias resistors; supervisor expander bit map and stereo channel-selection resistors agree	`checks/critical-pin-review.json`. Not a complete electrical signoff
Current assembly sources	All eight fresh populated STEP files loaded, valid and linked by hash in the reopened native FreeCAD assembly	`checks/current-assembly-sources.json`
Full nominal assembly	19 assemblies / 171 pair comparisons; no unexpected intersections	`checks/full-assembly-collisions.json`
Supports and key travel	36 posts at zero nominal centre/height offset; 108 key envelopes clear the frame through 4 mm travel	`checks/support-and-key-travel-check.json`
Accessories and interchange files	23 accessory checks clear the compared solids; 26 mechanical STEP files reopen as valid shapes	`checks/detailed-fit-check.json`
Revised AON power parts	Current power assembly and supplier maximum L2 envelope clear the compared case/module/cable allocations	`checks/aon-mechanical.json`
Firmware builds	QMK/RP2040 and ESP32 compiled; shipped sources/binaries and refreshed upstream archives match recorded hashes	Build logs, `firmware/binaries/manifest.json`, `checks/source-archive-check.json`

Check	Result and scope	Evidence
Actual-source logic fixtures	8 QMK and 18 ESP32 scenarios pass AddressSanitizer/UndefinedBehaviorSanitizer; hardware APIs mocked	`checks/logic-current/logic-tests.json`
Power/protocol fixtures	29 supervisor scenarios, 45 individually addressed I2C fault cases, 771 C/Python packet vectors and 400 corruption trials pass	`checks/firmware-current/firmware-tests.json`, `checks/power-matrix/power-fault-matrix.json`
Software boundary fixtures	24 tests pass, including isolated pseudo-terminal framing, GPIO reset/cleanup, private USB gadget configuration and mocked Bluetooth failures	`checks/software-current.log`
Browser UI	Prior successful browser simulation retained only after byte-for-byte confirmation that its UI source is unchanged	`checks/retained-browser-evidence.json`. Historical simulation, not a new target run
ODS workbook	Three sheets, 409 formulas, 829 positions; totals and visible missing-quote cells reconcile	`checks/ods-check.json`, `checks/workbook-check.json`
Assembly drawings	All 16 front/back views fit their pages without clipping; refreshed from the current boards	`checks/assembly-pdf-check.json`
Audit integrity	Current board/export/model/source/report hashes reconciled; Gerber ZIP CRC/content checked	`checks/audit-integrity.json`. Does not authorize fabrication

Mechanical scope

The only waived full-assembly intersections are two known CM4 connector mating regions, totaling approximately 1.36227 cubic millimetres with the current laminate thickness. Their bounds are recorded individually. No unrelated overlap is suppressed.

Some models are simplified maximum-dimension packages or purchased-module envelopes. The CM4 supplier model includes the previously documented conservative repair of one invalid solid. Valid nominal geometry is not physical metrology. Own-board mating plug envelopes are excluded where they intentionally contain their connector body. Formed cables, latches, purchased part tolerances, screw engagement and manufacturing shrinkage need real samples.

The current native assembly uses all eight refreshed populated board exports. Mechanical part STEP files remain the same case geometry and reopen successfully. Custom heatsink performance, battery retention, display-ribbon folds and connector insertion forces have not been measured.

Software and hardware scope

The tests use real project source against controlled fixtures. They do not exercise actual regulators, radios, USB links, audio devices or the target Linux system. LeakSanitizer was disabled because this sandbox does not support its ptrace requirement; no leak-test result is claimed. Build success and source hashes do not prove hardware operation.

All 26 physical/procurement acceptance rows remain NOT TESTED. Stackup/SI signoff, capacitor/ripple margins, USB input startup/inrush, battery documentation and complete electrical review remain open before an approved build. No owner PC installation, device flashing, radio pairing, service activation or host audio changes were performed.